

Iliass Elyaakoubi Benssaleh – Data Scientist

Analytical and purposeful Data Scientist with experience in Data Science and Machine Learning, specialising in machine learning, computer vision, and data-driven problem solving. Skilled in developing full-stack solutions, optimising workflows, and delivering AI-powered tools that drive measurable business value. Experienced in stakeholder engagement, mentoring, and translating complex requirements into scalable solutions.

WORK EXPERIENCE

Nov 2024 – Present | Data Scientist – Rebel Energy Labs, London

Summary: Work within the Data Science team on predictive modelling engagements supporting energy supply operations in a rapidly growing scale-up. Responsible for model research, development, and validation.

- **Computer Vision for Conservation:** Partnered with Rebel Restoration to develop a fish population monitoring system from underwater footage as part of a seagrass rewilding initiative. Trained object detection models (YOLOv8/YOLOv11) integrated with ByteTrack to explore automated species identification and tracking. The project demonstrated potential applications of computer vision in biodiversity restoration.
- **Exception Management System:** Designed and deployed a Django-based enterprise exception & data quality platform that consolidated billing errors, customer data mismatches, and operational anomalies into a real-time dashboard (TailwindCSS, Plotly). Integrated 40+ SQL-driven data quality tests, enabling closure of 67,000+ quality issues within 2 months and giving operations instant visibility to fix root causes. Post-launch, open exceptions fell by 20% through improved transparency and ownership.
- **Business-Facing Data Analysis:** Collaborated with the Supply team to deliver ad-hoc analyses and reporting pipelines via SQL and PySpark, working within Confluence (documentation) and Jira (task management). Produced actionable insights supporting operational decisions, including validation work on the commission model, and strengthened engagement between data and business teams.
- **Payment Allocation Model:** Designed a framework to automate Rebel's manual bulk billing payment allocation process, reducing reconciliation time and enhancing financial accuracy. Delivered a proposal to the Finance team outlining how SQL-driven rules could streamline allocation workflows, providing a blueprint for future implementation.
- **Regulations LLM:** Built a prototype AI compliance assistant using Llama 3, LangChain, and retrieval-augmented generation (RAG), presented through a Streamlit web app. The system let non-specialists, including the Legal team, ask natural language questions about Ofgem regulations and instantly receive clear answers. This proof-of-concept demonstrated how advanced AI tools can simplify access to complex regulatory information across the business.
- **Mentoring:** Supervised and supported two data analysis interns contributing to Supply team requests. Reviewed SQL deliverables, provided technical feedback, and offered guidance on analytical approaches, ensuring quality and consistency in outputs while strengthening intern capability in practical data analysis.
- **Anomaly Detection Research:** Conducted research and implemented techniques for identifying irregularities in 19 million meter readings and billing transactions, performing benchmarking on Isolation Forest, Local Outlier Factor (LOF), autoencoders, and clustering methodologies. Applied an ensemble approach for anomaly detection, highlighting LOF as a particularly effective method for identifying unusual consumption patterns and operational errors in large-scale energy data.
- **Commission model:** Conducted a forensic reconstruction of Rebel's commission framework during administration proceedings. Applied SQL and Python to rebuild receivables models, collaborating within a four-person data team to support external administrators. The analysis uncovered a £6M+ discrepancy, providing critical evidence of system gaps and informing creditor reporting.

Apr – Jul 2024 | Computer Vision Intern – Checklens, Cyprus

Summary: Worked within the Computer Vision team on real-time self-checkout systems, focusing on model training, data annotation, and evaluation to support object detection research.

- Trained and evaluated YOLOv8 models on video frames from self-checkout systems, combining pre-labelled data with manually annotated items to improve model understanding.
- Conducted firsthand data annotation and model performance assessment, gaining practical experience with object detection workflows and ML evaluation techniques.

Oct 2019 | Web Technology Intern – Softwire, London

Summary: Worked within the web development team to build a prototype interactive quiz application, focusing on front-end integration and API-based automated grading.

- Developed a dynamic quiz interface using HTML, CSS, and JavaScript, allowing users to answer questions and receive immediate feedback.
- Integrated API calls to load questions and automatically grade responses, enabling a fully functional prototype for testing and demonstration.

EDUCATION

2023 – 2024 | MSc Artificial Intelligence (Distinction) – Queen Mary University of London

- Key modules: Machine Learning (75.5%), Neural Networks & NLP (95.2%), Artificial Intelligence (93%), Computer Vision (72.3%)

2020 – 2023 | BSc Computer Science (First Class Honours) – Queen Mary University of London

- Key modules: Algorithms & Data Structures (93.2%), Software Engineering (72.4%), Deep Learning & Neural Networks (81%)

SKILLS

- **Programming & Development:** Python, Java, HTML, JavaScript, Django, React, Figma, Git, Azure DevOps
- **Machine Learning & AI:** PyTorch, TensorFlow, Keras, LangChain, RAG, Object Detection (YOLOv8/YOLOv11, ByteTrack, DeepSORT), Neural Networks, Anomaly Detection
- **Data Analysis & Engineering:** SQL, DBT, PySpark, Polars, Data Modelling, ETL Pipelines, Data Visualisation (Plotly, Matplotlib)
- **Visualization & Reporting:** Plotly, Matplotlib, Data Storytelling
- **Tools & Documentation:** Confluence, Jira, Microsoft Office

CERTIFICATIONS

- Udemy – Python Django, the practical guide
- Coursera – Machine Learning Specialization
- Coursera – Deep Learning Specialization